2950 Niles Road, St. Joseph, MI 49085-9659, USA
269.429.0300 fax 269.429.3852 hq@asabe.org www.asabe.org

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Commercial Application of Pyrolysis Technology in Agriculture

A.J. (Sandy) Marshall[2], Ping F. Wu[1]; Sang-Hun Mun[1]; Charles Lalonde[2]

1. Ontario Ministry of Agriculture and Food, Guelph, ON, Canada.
2. Ontario Federation of Agriculture, Guelph, ON, Canada.

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Abstract.

The conversion of agricultural, food, biofuel and forestry residues into value-added products via pyrolysis represents many advantages. Bio-char is created during pyrolysis and to a minor extent from gasification and imperfect combustion processes. Bio-char can be applied to soil as a means to improve soil health, to filter and retain nutrients from percolating soil water. Two business cases have been developed to provide an indication of the state of commercialization of pyrolysis technology in agriculture. Manpower costs constitute the most significant part of the operating costs of a stationary pyrolysis plant. Increasing the throughput of the pyrolysis process with constant manpower improves the business case. Mobile pyrolysis plants may successfully reduce raw material and product transportation costs but are subject to increased labour and set-up costs, depending on how often and how far the unit are moved. Ultimately, the probability of success of both mobile and stationary pyrolysis business models will heavily depend on the steady availability of feedstock and on mitigating raw material costs. Improvements in pyrolysis technologies can further increase yields and process energy efficiency increasing the probability of a strong business model.

Keywords. bioenergy, agricultural products, biomass, soils, biomaterial.

INTRODUCTION

Biomass is produced by photosynthesis and captures atmospheric CO_2. This carbon is sequestered into the biomass but is normally released back into the environment by breakdown of the biomass through normal aerobic and anaerobic processes such as combustion, composting, rotting and fermentation. Generally CO_2 is the primary gas generated through these processes. However in anaerobic processes, methane is also generated and has a greenhouse gas impact over 100 years, which is 25 times higher than that of CO_2 (Wikimedia, 2013). Pyrolysis is a process in which the carbon content of the biomass is segregated into three components: non-condensable gases, bio-oil and bio-char. In most cases, NCG and bio-oil are used as energy sources and this portion of the carbon is released again back into the environment primarily as CO2. Bio-char, when used for soil amendment or other agricultural applications, provides the opportunity for the carbon to be sequestered for long periods of time. According to the Alberta Biochar initiative, 50 wt% of the carbon in the biomass can be concentrated in the bio-char and can be sequestered (Alberta Biochar Initiative, 2013). This life cycle is shown in Figure 1. The amount of carbon sequestered from the biomass depends significantly upon the yield of bio-char from the biomass. In the case of fast pyrolysis, bio-char yields are normally ca. 20 wt%. The ability to sequester carbon will increase with increasing bio-char yield. Auger-style pyrolysis provides bio-char yields of up to 50 wt% thereby enhancing the carbon sequestration potential. More work is required to better understand the ability of bio-char to sequester carbon in soil for the longer term.

Companies such as Walmart are driving sustainability through their procurement strategy. There is a great potential to place bio-char into consumer applications such as organic fertilizers, compost blends and other horticultural applications.

Westerhof outlines the effects of feedstock water content on the pyrolysis and condensation conditions (Westerhof, et al., 2007). Above 20 wt%, excess water in the feedstock simply vaporizes in the reactor and then creates challenges and product loss in the condensation system due to the sweeping water vapour pressure. Therefore, if excess water is present, one would have to operate the reactor to obtain more heat generation. In order to accomplish this, the pyrolysis reactor must be operated at higher reactor temperatures to create more thermal cracking leading to the production of extra pyrolysis product gas. This gas can be recycled and combusted to generate the extra heat to compensate for the higher water content.. As a result of operating at a higher temperature, bio-oil and bio-char yields are reduced and CO_2 emissions increase.

If sacrificing bio-oil and bio-char yields is not preferred, then excess heat for water vaporization can be provided by fossil fuels such as propane or natural gas to the reactor furnace. Unfortunately, this will result in increased emissions. It is estimated that for every 1% increase in feedstock moisture content (above 20 wt%), the MPS units (processing dry tonnes of biomass per day) would emit an extra 18-20 kg of CO_2 per day. As a result, ideal organic feedstock for pyrolysis units should contain less than 20 wt% moisture content.

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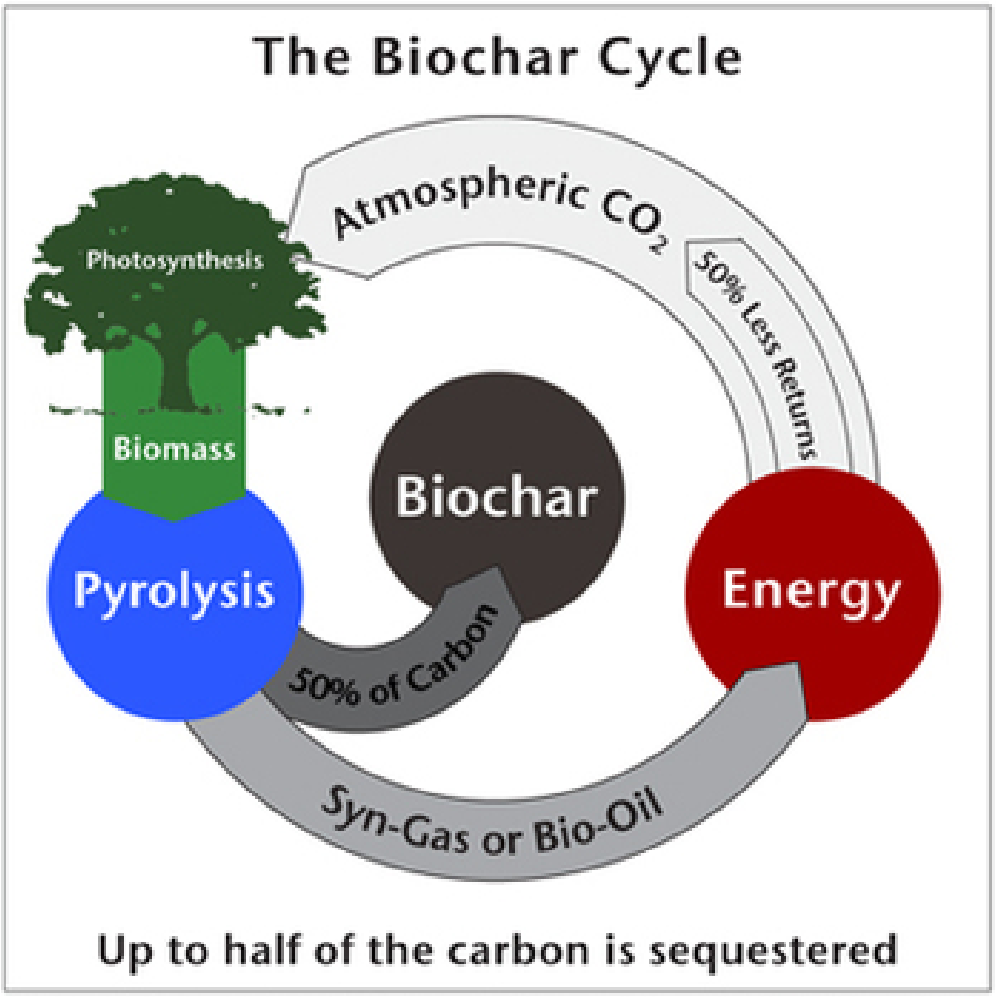


Figure 1 - The Bio-char Cycle of Carbon Sequestration (Alberta Biochar Initiative, 2013)

Agricultural Feedstock for Pyrolysis

There are wide array of agricultural biomass that can be considered as potential feedstock for commercial pyrolysis applications. Each source has its own set of issues and opportunities that need to be considered when assessing its potential for commercial use with pyrolysis technology. This section of the report will review the current availability and state of knowledge regarding the most prominent sources of agricultural biomass.

Corn Stover and Wheat Straw

Corn stover can be sustainably removed from farmland in Ontario. Based on Aung and Lalonde's work within the four county area of Lambton, Huron, Middlesex and Chatham-Kent, the average sustainable corn stover harvest rate is approximately 1.2 tonnes per acre. On a 100% dry matter basis, this equates to approximately 1.0 tonne per acre (Western Sarnia-Lambton Research Park,

2012). This is based on the assumptions that conservation tillage is used and that corn is grown in a rotation with soybeans and wheat. This is a typical rotation for many farms in this region. From 2003 to 2012, the average amount of corn stover that could be harvested sustainably in this region was over 500,000 dry tonnes per year (Western Sarnia-Lambton Research Park, 2012).

Wheat straw is also available Southwestern Ontario in large amounts. Wheat straw can be sustainably harvested at 1.2 tonnes per acre (15% moisture). The amount of wheat straw that could be harvested sustainably in this region was over 350,000 dry tonnes per year (Western Sarnia-Lambton Research Park, 2012).

Both biomass sources are produced once annually (summer for straw and fall for corn stover) which must be considered in the logistics of the pyrolysis processing. Either of these biomass sources would need to be stored to spread out their processing throughout the year or they would need to be processed on a campaign basis following harvest. These considerations will have an impact on the size and mobility of any pyrolysis processing plant.

Bale handling and biomass feed preparation (e.g. grinding) would need to be considered as part of the pyrolysis processing of these biomass sources.

Harvest costs for corn stover where recently studied and summarized as part of a larger corn stover to bioproducts study (Duffy & Marchand, 2013). From this work, stover costs based on harvest costs (discbine, raking and baling, stacking) averaged $66 per dry tonne over the last ten years. Nutrient value in the corn stover would add, on average over the same period, $9 per dry tonne to the cost of the stover. This value would be minimized if the bio-char from the pyrolysis was returned to the farm land for soil amendment.

There continues to be much discussion about the potential detrimental effects of excess corn stover on the subsequent year's crop yields as corn yields continue to climb higher. At some point in time, corn stover may need to be removed from the fields to minimize this impact. At this point in time, the corn stover would require processing in some fashion and pyrolysis could be a solution.

Wheat straw has alternative uses in livestock feed rations and as bedding. Based on the annual fair market value from 2003 to 2012 reported by Agricorp, the value of straw ranged from $89 to $149 per tonne with an average of $108 per tonne (Duffy & Marchand, 2013).

As noted above, there is an abundant supply of corn stover that can be sustainably removed from agricultural lands in Ontario. Corn stover is made up of cellulose, hemicellulose and lignin. It is a typical agricultural biomass and has a composition which is representative of other agricultural biomass such as wheat straw and purpose grown crops such as miscanthus and switchgrass. Pyrolysis products from wheat straw biomass sources are similar to those from corn stover (Demirbas, 2004 & Pütün, 2002). Fast pyrolysis of corn cobs and corn stover has also been studied in detail (Mullen, 2010).

Bio-oil from corn stover does not contain significant quantities of known chemicals with unique bioactive or pharmaceutical properties. As such, bio-oil from corn stover will be limited to use as a fuel source in commercial energy applications.

A mobile fast pyrolysis unit could be used in this application as energy densification is needed to minimize logistics costs. Collection and transportation of the bio-oil to a centralized power generation facility could be envisioned. Bio-char produced remotely could be used as soil amendment thereby ensuring that the minerals contained in the biomass are returned to the farm. However, bio-char from corn stover residues has not yet been sufficiently studied to determine its quality and establish a value as a soil amendment above an accepted baseline value of $250 per tonne.

Energy Crops

Crops such as miscanthus, energy sorghum, tall prairie grass and switchgrass have the potential to be used as biomass feedstock for pyrolysis. As noted above, these energy crops consist of the cellulose, hemicellulose and lignin and will produce similar bio-oil and bio-char as corn stover and wheat straw (Palma, Richardson, & Roberson, 2011).

A study conducted for the Ontario Federation of Agriculture looked at the price for purpose-grown crops that would be needed to provide a net return to the farmer of $100 per acre (Western Sarnia-Lambton Research Park, 2012). The acceptable price of purpose-grown biomass at the farm gate in Ontario ranges from $104 per dry tonne to $148 per dry tonne, based upon industry-based inputs, average production and cost estimates outlined in the study. Of the purpose-grown crops identified, miscanthus offers the lowest production cost due to its comparatively high yield. Based on the analysis presented in this study, the acceptable price of miscanthus bales at the farm gate would be $104 per dry tonne to provide a comparable margin with conventional cash crops. The establishment cost of miscanthus including the fixed costs is $1179 per acre, but can vary widely from farm to farm as producers are just learning how to grow the crop. A decrease in the establishment cost by $300 to $879 per acre will reduce the acceptable price of miscanthus bales at farm gate by approximately $7 per dry tonne. The acceptable price of switchgrass bales at the farm gate would be $135 per dry tonne. The acceptable price of biomass bales at the farm gate for Tall Grass Prairie (TGP) and sorghum are $148 per dry tonne and $103 per dry tonne, respectively. Although TGP offers the maximum environmental benefits, information on fuel quality of the mixed biomass is limited. The higher establishment cost and the relatively lower yield could be issues. The higher moisture content of current sorghum species at harvest is also an issue in using this feedstock for pyrolysis.

Jerusalem Artichoke

Jerusalem artichoke has been investigated for its potential as a biomass source for pyrolysis to produce biofuels and biochemicals that can be used for food additives, flavours, and pharmaceuticals (Encinar, et al., 2009 & Scachetto, 2012). The study concluded that the Jerusalem artichoke plant can be pyrolysed to produce bio-oil containing chemicals such as acetic acid, furfural, acetone and carbonyl derivatives in reasonably high concentrations. Other interesting biochemicals were found in lesser concentrations. In particular, flavouring potential was found in these bio-oils. Isolation and removal of these chemicals from the bio-oil is not economically viable at this time.

Energy recovery is possible from the bio-char and the non-condensable gas where there is a high carbon and hydrogen concentration. The bio-oil generally had lower heating values due to the higher concentration of water.

Extraction of soluble carbohydrates such as glucose, fructose and inulin from the Jerusalem artichoke roots prior to pyrolysis, enhances the pyrolytic yields without compromising the yield of biochemicals. However, it is not of commercial relevance as Jerusalem artichoke is not widely grown and the economics of pyrolysis are not proven.

Dried Distillers Grains (DDG)

Dried distillers grains are produced from the production of grain ethanol using corn. DDG is produced continuously throughout the year in the ethanol production facilities and is free-flowing as a dried material minimizing the feed preparation for the pyrolysis process. There are a number of plants in Ontario such as Greenfield Ethanol, Suncor Ethanol and IGPC. Potential production of

DDG is estimated at 180,000 tonnes per year based on the combined capacities of the ethanol production facilities operating in the Province of Ontario (Western Sarnia-Lambton Research Park, 2012).

Dried distillers grains are normally sold as animal feed and the price is set in the market based on supply and demand. With the significant growth in ethanol production over the last decade, the market for DDG has been oversupplied resulting in lower prices for this material.

A wholesale price of by-product DDG of $180 per tonne has been used in previous economic assessments of alternative technologies (Western Sarnia-Lambton Research Park, 2012).

Products produced from the pyrolysis of DDG have been characterized (Xu, 2010; Wang, et al., 2012; Lei, et al., 2012). The products of DDG pyrolysis products are similar in composition and heat content to those from other agricultural sources such as corn stover and wheat straw.

Mushroom Residues

There are 300 million tonnes of wet mushroom biomass residue (65% moisture content) produced in Canada each year of which approximately 60% is produced in Ontario. The biomass is the residual growing media after the mushroom harvest. The growing media is initially a blend primarily made of various peat materials, straw and chicken litter. This material creates significant issues for the mushroom industry due to the sheer volume produced. This volume of wet biomass is concentrated on approximately 36 farms in Ontario, most of which are geographically located within Middlesex County and the Niagara region.

Due to the significant volume and inability to reuse the material for further mushroom production, the biomass represents a disposal cost to the farmers. Currently, the disposal costs are estimated at $10 per wet tonne of biomass.

Whitecrest Mushrooms Ltd has been working with Canadian Carbon Solutions (CCS) to develop a process for the pyrolysis of mushroom biomass. In this process, the biomass must first be dried to 10% residual moisture content before being processed in CCS's unique auger-style pyrolysis unit to produce a bio-char with good surface properties. The bio-char has been used for soil amendment in agricultural applications and has a projected commercial value of $430 per tonne.

CCS's unique process of sequestering carbon in bio-char is different from all other technologies. The process raises feedstock temperature up to the thermochemical equilibrium point, converting the calorific value of the feedstock to a bio-char with high carbon purity. The pilot unit is capable of producing a bio-char of consistent quality. CCS has proven that bio-char properties can be controlled in the pilot unit through process adjustments. This allows CCS to operate with various feedstocks and meet diverse customer requirements.

A pyrolysis plant would include the following:

- biomass and bio-char handling equipment
- a steam heated rotary dryer
- a CCS auger-style pyrolysis unit for production of bio-char
- a low pressure steam boiler for combustion of the bio-gas and bio-oil
- an integrated energy management system to minimize energy consumption

Greenhouse Vegetable Growers

Vegetable greenhouse operations produce primarily tomatoes and peppers in Ontario. According to the Greenhouse Growers of Ontario, there are 2,100 acres of greenhouses where vegetables are grown in Ontario. These operations are concentrated in the Essex region where over 75% of the acreage is grown with most of the remaining acreage grown in Huron, Kent, Lambton and Niagara Counties.
The organic biomass from vegetable greenhouses consists of vines and culls from the plants. There are lesser amounts of growing media in the biomass which is made up primarily of rockwool (silica) and washed coconut husks. The greenhouses are cleared once per year with between 20 to 30 tonnes of wet biomass removed per acre of greenhouse. Total biomass available from this source is approximately 50,000 tonnes per year. The moisture content of the biomass is unknown but is expected to be in the range of 50-60 wt%.

Currently, the biomass is removed from the greenhouse sites by a third party waste handling contractor. This contractor is centrally located in Essex County and removes the biomass from the sites, then separates the plastic for recycling prior to processing the biomass. The contractor charges a tipping fee in the range of $36 - $40 per wet tonne.

As stated above, the biomass is produced once per year. This agricultural residue would need to be processed on a campaign basis within a short period of time due to its high moisture content. It is likely that the material would need to be removed to a central processing location as is currently done by the third party contractor unless there was a financial incentive for the greenhouse operator to accept the inconvenience of on-site processing.

Greenhouse Flower Growers

According to Flowers (Canada) Ontario, there are very little organic biomass residues from their industry. Most growers in Ontario are producing potted plants for the spring season and very little biomass remains after the sale of the potted plants. Cut flowers are not grown in Ontario to any extent.

Tobacco Residues

Bio-oil from tomato and tobacco residues has shown bioactive properties. Unfortunately, the bioactive chemical species are not yet clearly identified so that work can be done to isolate them from the bio-oil. Availability of tobacco residues is currently unknown.

Processed Vegetable Residues

Field tomatoes, corns, peas and beans are grown in Ontario for the processed vegetable industry. The majority of this production is concentrated in the Southern Ontario region between Strathroy and Leamington.

Information was not readily available for quantities and cost of corn, peas and bean residues. However, the majority of processing plant residues from these crops would be corn cobs and husks. These residues would have high moisture content (50-60 wt %) and would require pre-drying before pyrolysis. The products from pyrolysis of the dried biomass would be similar to the products from field corn stover and cobs.

The majority of field tomatoes are processed by three companies in Ontario. Collectively, 500,000 tonnes of fruit are processed per year. Approximately 5,000-7,000 wet tonnes (60% residual moisture) of tomato pomace are produced per year in Ontario. The pomace is made up of skins and

seeds and is produced during the 5 week harvest period each fall. It is sold currently for $10-$20 per wet tonne and is used in cat and dog food as a source of soluble fibre. Until such time as there are opportunities to extract high value chemicals, pharmaceuticals or pesticides, tomato pomace will continue to be used in cat and dog food.

Grape Skins and Grape Seed Residues

Grape skin and grape seed residues (6% moisture content) have been pyrolyzed at the pilot scale to produce bio-oil and bio-char (Xu, et al., 2009; Demiral & Ayan, 2011). Under fast pyrolysis, the yield of bio-oil is lower than that seen from other agricultural biomass such as corn stover. Bio-char yields were higher as a result of the lower bio-oil yields. It was also noted that bio-char yields continued to increase as the residence time increased to 20 seconds and the pyrolysis temperature increased to 500°C (Xu, et al., 2009).

The heat of pyrolysis was also measured and compared with the available energy in the non-condensable gas (NCG) stream. It was determined that there was insufficient energy in the NCG to fully sustain the pyrolysis reaction and that a secondary energy sources such as bio-oil or natural gas would be required for a commercial application.
An approximate quantitative analysis of the bio-oil product components was done and the study indicated that the concentration of valuable chemicals is not high enough to justify their extraction.

Due to the tendency for higher bio-char yields, the use of grape skins and grape seeds for specialty bio-char applications could be of interest.

Meat and Bone Meal (MBM)

Experiments have proven that it is possible to feed dried meat waste and MBM (5 – 6 wt% moisture content) continuously in a pilot scale fluidized bed pyrolysis unit under typical conditions (Cascarosa, et al., 2011; Cascarosa, Becker & others, 2011; Burruti, et al., 2012).

The bio-oils produced in the MBM pyrolysis process have a dark brown and homogeneous colour and a strong pungent odour. The average heating value of the bio-oil obtained was significantly higher than that of the bio-oil obtained from cellulosic biomass. However due to their high nitrogen content (7% – 9%), these bio-oils could not have a direct use as a fuel (Cascarosa, et al., 2011). The bio-oils contain mainly fatty acids, fatty nitriles and some fatty acid esters.

The bio-char yields from the pyrolysis of meat waste and MBM range between 20–30 wt%. Not surprisingly, the ash content of the bio-char was very high and in the range of 63–77 wt%. The ash contains a high proportion of calcium in the form of carbonates and phosphates (Cascarosa, et al., 2011).

As noted here, the properties and composition of the bio-oil and bio-char from the pyrolysis of MBM are significantly different that those from other biomass sources. Further study is warranted to understand the commercial impact of these differences.
Currently, there are two types of MBM:

1. MBM that is suitable for animal consumption and is used in high protein animal feed. This product has a high market value
2. MBM that cannot be used for animal feed since it may be contaminated with disease such as Bovine Spongiform Encephalopathy (BSE). This material is incinerated at temperature above 850°C in order to meet the regulatory requirements for destroying reinfectivity of disease. Pyrolysis does not meet the threshold temperatures required for this application.

Consequently, the pyrolysis of MBM is not seen as providing commercial value.

Chicken Litter

Chicken litter is a high nitrogen content (6 wt% nitrogen) biomass material which is rich in cellulose as sawdust is a major component used as bedding material (Schnitzer, et al., 2007). Dried chicken litter has been processed in a pilot scale fast pyrolysis auger-style unit. Approximate yields of 10% non-condensable gases, 63% bio-oil and 27% bio-char were obtained from this process (Monreal & Schnizer, 2011; Schnitzer, et al., 2007; Kazi, et al., 2011).

In a first study, the raw bio-oil contained 6.3% nitrogen and 23.3% oxygen (Monreal & Schnizer, 2011). Later work obtained bio-oil with nitrogen contents of 7.4 and 12% (Schnitzer, et al., 2007). As noted with the bio-oil from MBM, due to the high nitrogen content, this bio-oil does not have a direct use as a fuel. However, through further chemical processing, the authors were able to produce a refined bio-oil with less than 2% residual oxygen content and 0.2% nitrogen content. Unfortunately, the yield was only 10 wt% based on the bio-oil processed. Although this approach is interesting, the status of this work is still experimental and not yet of commercial significance.

An analysis of the bio-char resembled that of the initial chicken litter except that the concentration of carbohydrates was lower (Schnitzer, et al., 2007).

Other Raw Material Sources

The breadth of work conducted on pyrolysis of carbon sources is endless. For the sake of completeness, references for various other raw material sources are list here:

- Sawdust (Demirbas, 2004; Bridgwater, et al., 2001; Xu, 2010; Czernik & Bridgwater, 2004; Bridgwater, 2012; Radlein & Piskorz, 1997; Meier, et al., 2013; Elliott, et al., 2012; Grigiante, et al., 2010).
- Lignin (Palmisano, et al., 2011; Pielhop, et al., 2011; Zhang, et al., 2011; Ingram, et al., 2008; Tapin-lingua, et al., 2011; Gosselink, et al., 2011).
- Coffee Grounds (Bedmutha, et al., 2011 & Bok, et al., 2012).
- Wastewater Treatment Sludge (Grigiante, et al., 2010; Manara & Zabaniotou, 2012).
- Polymer and Tires (Grigiante, et al., 2010; Quek & Balasubramanian, 2013)

Business Cases for Commercial Application of Pyrolysis

Two business cases have been developed to provide an indication of the state of commercialization of pyrolysis technology in agriculture. The cases are described as follows:

1. Mobile fast pyrolysis of biomass (20 wt% moisture) without pre-drying. The bio-oil is collected and transported to a centralized combined heat and power (CHP) generating facility. The mobile unit is energy self-sustained by using the non-condensable gases from pyrolysis. The CHP unit generates steam for the production of renewable power (FIT price 13 cents/kWh). The residual steam is used to displace natural gas as a heat source for a secondary purpose. Natural gas is priced at 30 cents/m^3. The bio-char produced is of lower technical quality and is valued at the lower end of the price range ($250 per tonne). Corn stover will be used as the biomass source for this business case.
2. Integrated auger-style slow pyrolysis of biomass (10 wt% moisture content) with pre-drying in a steam heated rotary dryer. The non-condensable gases and bio-oil vapours are directly burned in a steam boiler which provides the heat source to dry the wet biomass prior to pyrolysis. To close the energy balance, natural gas is used to direct heat the auger-style pyrolysis unit and to supplement steam production in the boiler. Natural gas is priced at 30¢/m^3. The bio-char produced is of higher technical quality and is valued in the middle of the price range ($430 per dry tonne). Mushroom residue (65 wt% moisture content) will be used as the biomass source for this business case.

Case 1: Mobile Fast Pyrolysis of Corn Stover

Operating Conditions for conversion of 3,600 dry tonnes per year of biomass

Operating conditions were developed for the mobile pyrolysis unit scenario using the following assumptions:

1) Pyrolysis unit has a capacity of 10 dry tonnes per day corn stover. The corn stover processed has a moisture content of 20 wt%.
2) Pyrolysis unit is operated on a continuous basis, 24 hours per day. Unit utilization rate is 95% (350 operating days per year). Corn stover processed is 3,600 dry tonnes per year (4,500 wet tonnes per year).
3) The product slate from the fast pyrolysis of corn stover is described in **Fig. 2:**
   a) 21.9 wt% non-condensable gases (higher heat content (HHV): 6 MJ/kg)
   b) 61.6 wt% bio-oil (higher heat content (HHV): 22.1 MJ/kg “as is”)
   c) 17.0 wt% bio-char (market value: $250 per dry tonne)

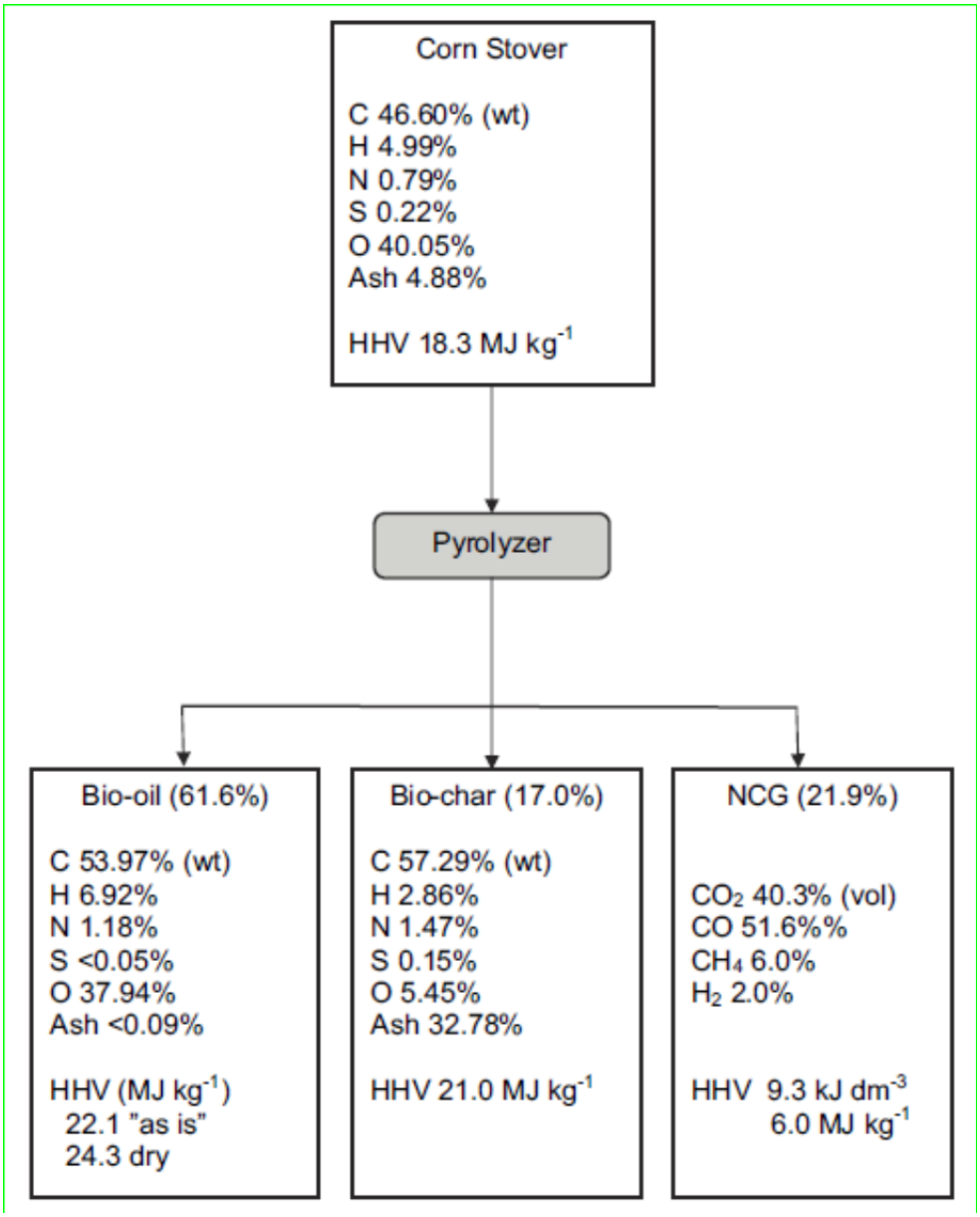


Figure 2: Product slate from fast pyrolysis of corn stover (Mullen, 2010)

Operating Costs

Operating costs were developed using the following assumptions:

1) Corn stover is delivered to the processing location at a price of $60 per dry tonne (20 wt% moisture content). Raw material costs have been determined at $337,500 per year
2) Two persons are required to operate the unit at all times. These employees will receive an hourly wage of $20 with overhead costs of 20%. Personnel costs have been determined at $403,000 per year
3) Bio-oil produced is collected, transferred and stored in a quad-axle semi-trailer. The trailer holds 40 m3 of bio-oil but is restricted to 34 tonnes for transport. It will be transported at its weight limit to a central CHP unit within 100 km. The trailer rental costs are included in the variable cost of transportation. The cost of transport is $966 per load of bio-oil (34 tonnes per load). Cost of transportation has been determined at $73,000 per year

4) Maintenance costs are set at 2% of capital costs. The costs of the mobile unit is estimated to be $1.5 million and the cost has been determined at $30,000 per year
5) Energy costs are minimal as the non-condensable gases and entrained bio-oil from the pyrolysis are used to provide an energy self-sustaining operation.
6) Other costs were determined at $20,000 per year. This category covers miscellaneous costs such as environmental, safety and logistics costs.

Energy Balance of Mobile Pyrolysis Unit and CHP Plant

The energy balance around the mobile pyrolysis unit will define the quantity of bio-oil that is available for shipment to a CHP generation plant for conversion into steam and power. The following assumptions have been used to determine the volume and energy value of the bio-oil produced by the mobile pyrolysis unit:

1) The quantities and the higher heat value (HHV) of the non-condensable gases and the bio-oil are summarized above in Section 6.1.1.3
2) The heat of pyrolysis describes the energy required to pyrolyze the biomass within the pyrolysis unit. This quantity of energy is process and biomass specific. A value of 1.75 MJ/kg is used and based on input from a study (Berruti, et al., 2013).
3) Based on the pyrolysis system mass balance, HHVs (NCG and bio-oil) and the heat of pyrolysis, 2,150 tonnes per year of bio-oil will be transported to the CHP plant

Revenue Streams from conversion of 3,600 dry tonnes per year of biomass

Revenue generated from this mobile pyrolysis unit was determined using the following assumptions:

1) Boiler/steam turbine CHP system performance data was used to determine the quantities of steam and electricity that would be produced from the bio-oil provided from US EPA (US EPA, 2013). A 500 kW back pressure CHP system was used to process the bio-oil. Based on the reference used, this unit has a CHP electric efficiency of 6.4% and a total CHP efficiency of 79.6%. From the energy balance, 98 kW of electricity would be produced in the CHP unit. Using an electricity price of 13¢/kWh (FIT), the value of this electricity is $118,000 per year. Similarly, 89,000 MJ/day of steam would also be generated from the bio-oil. The value of displacing natural gas to produce this quantity of steam is $272,000 using a natural gas price of 30¢/m^3)
2) Bio-char will be collected continuously, quenched to avoid spontaneous combustion and stored in drums. Bio-char is valued at $250 per dry tonne ($153,000 per year) based on the details described in Section 3.3

Outcome of Business Case Analysis

The business case for a mobile pyrolysis unit processing 10 dry tonnes per day (3,600 dry tonnes per year) of corn stover is summarized in Table 1.

A second business case was developed using the same assumptions and a throughput rate of 23 dry tonnes per day (8,000 dry tonnes per year). The result from this business case is also summarized in Table 1.

	Biomass Throughput Rate (dry tonnes)	
Operating Costs	**10 Tonne/day**	**23 Tonne/day**
Raw Materials	$337,500	$750,000
Personnel	$403,000	$403,000
Logistics	$73,000	$137,000
Maintenance	$30,000	$30,000
Energy	-	-
Other	$20,000	$20,000
TOTAL COST	**$863,500**	**$1,340,000**
Revenue		
Electricity	$118,000	$263,000
Steam	$272,000	$604,000
Bio-char	$153,000	$340,000
TOTAL REVENUE	**$543,000**	**$1,207,000**
CASH FLOW	**($320,500)**	**($133,000)**

Table 1 - Business Cases for Mobile Pyrolysis Units Converting Corn Stover

Cash flow is negative in both of these two cases and the 10 tonnes per day (4,000 tonne/year) case can't become cash flow positive using any reasonable assumptions.
A sensitivity analysis on cash flow was conducted on the 23 tonnes per day (10,000 tonnes per year) capacity case using the following variables:

- Price of corn stover per dry tonne. When keeping the bio-char and renewable energy prices constant, the price of corn stover (20 wt% moisture content) would need to be reduced from $60 per dry tonne to $50 to reach a breakeven cash flow excluding the cost of capital
- Price of bio-char per dry tonne. Keeping corn stover and renewable energy prices constant, the price of bio-char would need to increase to $350 per dry tonne to reach a breakeven cash flow excluding the cost of capital
- Price for renewable electricity under the FIT program (cents/kWh). Keeping corn stover and bio-char prices constant, the price for renewable energy would need to increase to 20 cents/kWh to reach a breakeven cash flow excluding the cost of capital

Since manpower costs constitute the most significant part of the operating costs, increasing the throughput of the pyrolysis process with constant manpower improves the business case. Consequently, a third scenario was created with a 30,000 tonne per year pyrolysis unit keeping the manpower requirements constant. Under these circumstances, a positive cash flow of $480,000 could be generated. It may be possible to generate a project payback at this scale.

A number of impacts could fundamentally change which would alter the conclusions made here. For example, if a dry biomass source became available at a lower cost (eg. if excess corn stover residue on the field became an issue, then removal would be required to ensure crop yield), then the economics of a large scale mobile pyrolysis unit would become more economical.

Outcomes from Similar Case Study

Palma outlined an interesting approach (Monte Carlo simulation) to predicting the potential for positive net present value (NPV) of investments and operations in mobile pyrolysis units in 2011 economic conditions using corn stover or energy sorghum in three locations in the USA (Texas, Illinois, Nebraska). The results of the simulation confirm a high sensitivity to yields, oil prices and the energy efficiency of the pyrolysis systems (Palma, et al., 2011)

The study uses a feedstock cost range of $60-$75US per dry tonne. There may be opportunities to obtain dry biomass at prices below these price points. However, most dry agricultural biomass identified in Ontario has a projected value equal to or higher than the costs used in this study.

The study assumed that the major income-generating product is a bio-oil equivalent to crude oil, which is transported to the nearest refinery, and sold at a 5% discount from the price of crude. The cost to upgrade the raw bio-oil to high quality bio-oil usable in an oil refinery was within the range of 20–40 cents per gallon and a subsidy of $1.00 per gallon of refined bio-oil was provided.

The bio-char value was determined based on the soil amendment value as a soil additive. According to this study, bio-char produced in 2011 can be sold for $24 per dry tonne and its value varies by feedstock. This value is significantly lower than that used in this study.

The results showed a low probability of economic success for all scenarios. Although this analysis was done using different assumptions, the general outcome from this assessment is consistent with those found in this study.

Case 2: Fixed Auger-style Pyrolysis of Mushroom Residue

Operating conditions for conversion of 3,200 wet tonnes per year of biomass

Operating conditions were developed for the integrated fixed auger-type pyrolysis unit. The auger-type pyrolysis unit is integrated into a plant with a steam heated rotary dryer, steam generation boiler and feeding systems using the following assumptions:

1) Pyrolysis unit has a capacity of 10 dry tonnes per day (3,500 dry tonnes per year) mushroom media residue. The mushroom residue processed has 65 wt% moisture content.
2) A steam heated rotary drier is used to reduce the moisture content of the mushroom media residue to 10 wt% moisture
3) The non-condensable gases and bio-oil vapours are combusted in a low pressure steam boiler and the steam is used to heat the rotary drier
4) The auger-style pyrolysis unit is operated on a campaign basis, 24 hours per day for extended periods. Unit utilization rate is 95%. The mushroom residue processed in this case is 3,200 wet tonnes per year (1,120 dry tonnes per year). Based on the capacity of the pyrolysis unit, there will only be sufficient feedstock to operate the unit for approximately 109 days per year.
5) The product slate from the slow pyrolysis of mushroom residue as described as follows (Palmisano, Briens, Lago, & Berruti, 2011)
   a) 10 wt% non-condensable gases (higher heat content (HHV): 1.8 MJ/kg)
   b) 40 wt% bio-oil (higher heat content (LHV): 13.6 MJ/kg)

c) 50 wt% bio-char (market value: $430 per dry tonne)

Operating Costs

Operating costs were developed using the following assumptions:

1) Mushroom media residue will be available at the processing location and will avoid a tipping fee of $10 per wet tonne (65 wt% moisture content). Raw material cost credit has been determined at $32,000 per year
2) Two persons are required to operate the unit at all times. These employees will only be required when the unit is in operation. These employees will receive an hourly wage of $20 with overhead costs of 20%. Personnel costs have been determined at $126,000 per year
3) Bio-oil vapour and non-condensable gases are injected directly into a low pressure steam boiler as fuel. Natural gas is used in the boiler to supplement the energy content of the pyrolysis products to close the energy balance. Natural gas is also used for direct heating of the pyrolysis auger-style unit. The boiler energy efficiency is 85%. The rotary drier efficiency was determined from "*Energy Manager Training*" to be 69% (Energy Manager Training, 2004). A natural gas price of 30¢/m^3 was used for this analysis. From this energy balance, the cost of natural gas has been determined to be $31,000 per year.
4) Maintenance costs are set at 2% of capital costs. The costs of the integrated pyrolysis unit is estimated to be $1.5 million and cost has been determined at $30,000 per year
5) Other costs were determined at $20,000 per year. This category covers miscellaneous costs such as environmental, safety and logistics costs.

Energy Balance of the Integrated Pyrolysis Unit

The energy balance around the integrated fixed auger-type pyrolysis unit determines the quantity of natural gas required. The following assumptions were used to determine the energy content of the non-condensable gases and the bio-oil produced by the auger-style pyrolysis unit:

1) The quantities and the higher heat value (HHV) of the non-condensable gases and the bio-oil are summarized above.
2) The heat of pyrolysis describes the energy required to pyrolyze the biomass within the pyrolysis unit. This quantity of energy is process and biomass specific. A value of 2.6 MJ/kg is used (Palmisano, et al., 2011). Revenue Streams for Conversion of 3,200 wet tonnes per year of Biomass.

The only revenue generated from the integrated pyrolysis unit is bio-char which is collected continuously, quenched to avoid spontaneous combustion. Bio-char is valued at $430 per dry tonne ($257,000 per year).

Outcome of Business Case Analysis

The business case for the integrated fixed auger-type pyrolysis unit processing 10 dry tonnes per day (3,200 wet tonnes per year) of mushroom media residue is summarized in Table 2.

Two additional business cases were developed using the same assumptions with the following changes:

- Operate the integrated unit at full capacity of 10,000 wet tonnes per year (340 days operation per year)
- Operate a larger integrated unit with a capacity of 30,000 wet tonnes per year

The results from these business cases are also summarized in Table 2.

	Biomass Throughput Rate (wet tonnes per year)		
Operating Costs	**3,200 T**	**10,000 T**	**30,000 T**
Raw Materials	($32,000)	($100,000)	($300,000)
Personnel	$126,000	$392,000	$392,000
Energy	$31,000	$98,000	$293,000
Maintenance	$30,000	$30,000	$30,000
Other	$20,000	$20,000	$20,000
TOTAL COST	**$175,000**	**$440,000**	**$435,000**
Revenue			
Bio-char	$257,000	$802,000	$2,405,000
TOTAL REVENUE	**$257,000**	**$802,000**	**$2,405,000**
CASH FLOW	**$82,000**	**$362,000**	**$1,970,000**

Table 2 - Business Case for Pyrolysis of Mushroom Residue

Cash flow was positive in all three scenarios. Cash flow was marginally positive for the 3,200 wet tonnes per year case and can be seen as the breakeven throughput case. Cash flow was sufficient from the 10,000 wet tonnes per year case to generate a simple payback period approaching 4 years. A sensitivity analysis on cash flow was conducted on the 10,000 tonnes capacity case using the following variables:

1) Removal of the tipping fee. For the 10,000 tonnes per year plant and keeping all other parameters constant, removal of the tipping fee would extend the simple payback period by almost 2 years
2) Price of bio-char per dry tonne. For the 10,000 tonnes per year plant and keeping all other parameters constant, the price of bio-char could be reduced to $250 per dry tonne before reaching the breakeven cash flow point excluding the cost of capital

Since manpower costs constitute the most significant proportion of the operating costs, increasing the throughput of the integrated pyrolysis plant with constant manpower will improve the business case. Consequently, the third business case was created with 30,000 wet tonnes per year capacity and constant manpower requirements. A positive cash flow of nearly $2 million per year was generated. If sufficient feedstock could be gathered and the auger-style pyrolysis unit could be scaled to this capacity, a very interesting business case could be generated.

Conclusion or Summary

The Conclusion or Summary section restates the major findings and suggests further research. It is the last main heading before the references.

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Put any acknowledgments, such as thanks to contributing individuals or organizations, here.

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Nomenclature

BFB	pyrolysis reactor type using a bubbling fluidized bed
BSE	bovine spongiform encephalopathy – mad cow disease
BTG	collaborative research program at Swansea University in England
°C	unit of measure of temperature – degrees Celsius
CaO	calcium oxide – a solid chemical compound

CFB	pyrolysis reactor type using a circulating fluidized bed
CHP	combined heat and power generation system
CO	carbon monoxide, a gaseous product from incomplete combustion
CO_2	carbon dioxide, a gaseous product from the combustion of fossil fuels
cp	unit of measure of viscosity - centipoise
CPB	Colorado Potato Beetle
DDG	dried distillers grain – the solid residual product from ethanol production
ESP	electrostatic precipitator
GC	gas chromatography
HPLC	high performance liquid chromatography
ICFAR	Institute for Chemicals and Fuels from Alternative Resources, Western University, London Ontario
IEA	International Energy Agency
IGPC	farm cooperative producing ethanol from cereal corn in Aylmer, Ontario
kPa	unit of measure of pressure – kilopascals
kWh	unit of measure of electrical energy – kilowatt-hour
LHV	measure of heat content – lower heating value
MAT	maximum allowable threshold
MBM	meat and bone meal
mg/ml	unit of measure of concentration – milligrams per milliliter
MJ/kg	unit of measure of heat content – megajoules per kilogram
m^2/g	unit of measure of surface area – square meters per gram of solid
MPS	mobile pyrolysis system
MWe	unit of measure of electrical energy – megawatt electricity
NCG	non-condensable gases generated during pyrolysis
NO_x	gases containing nitrogen and oxygen known for impacts on air quality
NREL	National Renewable Energy Laboratory
pH	unit of measure of acidity and basicity – logarithmic scale
ppm	unit of measure of concentration – part per million on a mass basis
SO_x	gases containing sulphur and oxygen known for impacts on air quality
TPG	tall prairie grass
USDOE	United States Department of Energy